Band-1 Hadamard Order 5956 — Miyamoto (Paley-I H_1488 + C_1490 over GF(1489)); Miyamoto-only Bonus (NOT in SPK); Gap 2052 Still OPEN

H_1488/C_1490/H_5956 Miyamoto PASS; Double audit PASS; Paley-I/II SKIP; 5956 Miyamoto-only (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_5924_Miyamoto`; `2026-09-15_Band1_Hadamard_5812_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_5956_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1488 and 5956 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with preferred Paley-I seed H_{1488} = Paley-I($q = 1487$) and symmetric Paley conference C_{1490} over **GF(**1489**)** (prime field, $q = 1489 \equiv 1 \pmod 4$). Double path for H_{1488} (Sylvester-double of Paley-I H_{744}, $q = 743$) is a **secondary/audit** construction (also verified). Classical SPK paths **FAIL**: Paley-I ($q = 5955 = 3{\cdot}5{\cdot}397$ not prime); Paley-II ($q = 2977 = 13{\cdot}229$ not prime); Kronecker (4×1489). Therefore order **5956 is a Miyamoto-only bonus (NOT in SPK)** (same honesty pattern as 5812/5476). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. `clay_closed = false`.

Abstract

Finite constructions of orders **1488** and **5956** via Miyamoto.

- H_{1488} = Paley-I ($q = 1487$) — preferred; Double audit of H_{744} ($q = 743$) also PASS
- C_{1490} = Paley conference over GF(1489) prime field
- H_{5956} = Miyamoto(H_{1488}, C_{1490})
- Paley-I / Paley-II for 5956: **SKIP**

SPK: 5956 Miyamoto-only bonus (NOT in SPK). Seed 1488 enters SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_5956.py`

Results. overall_pass=true; units 72/72; wall 174.6579s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 5956 = 4 \cdot 1489$; 1489 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1488} and C_{1490} (Paley conference over prime field 1489).
- Seed preferred: Paley-I $q = 1487$ (prime $\equiv 3 \pmod 4$).
- Double audit: half=744, $q = 743$ prime $\equiv 3 \pmod 4$ (PASS; secondary).
- SPK: Paley-I $q = 5955$ **FAIL**; Paley-II $q = 2977$ **FAIL**; Kronecker **FAIL** → **Miyamoto-only**.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_1488 Paley-I	Paley-I q=1487 (preferred)	0	2.815
H_744 Paley-I	Paley-I q=743 (Double audit half)	0	0.4645
H_1488 Double	Sylvester double of H_744 (audit)	0	1.4937
C_1490	Paley conf GF(1489) q=1489	0	2.4867
H_5956 Miyamoto	Miyamoto(H_1488, C_1490)	0	164.6777
H_5956 Paley-I	SKIP q=5955=3 · 5 · 397 not prime	—	—
H_5956 Paley-II	SKIP q=2977=13 · 229 not prime	—	—

SPK closure note

Seeds include prior SPK ships through **5924** (740/1480/5924) + **1452** + verified **1488/744**; **5956 NOT folded** (Miyamoto-only); **exclude** Miyamoto-only bonuses including **5812/5476/5284/5956**. Known count 1265. Next constructive gap **2052** (OPEN).

Quantity	Value
5956 in SPK	**False** (Miyamoto-only bonus)
1488 in SPK	**True** (Paley-I)
5924 in SPK	**True** (prior Paley-I)
5812 in SPK	**False** (prior Miyamoto-only)
Next constructive gap	**2052** (OPEN)

Next Miyamoto targets (after q=1489)

q	n=4q	need H	build notes
1493	5972	1492	[]
1549	6196	1548	['PaleyII(q=773)']
1553	6212	1552	['Kronecker(4x388)']

Honesty

clay_closed = false · GATE-04 LOCKED · Hadamard conjecture OPEN · gap 2052 OPEN · no RH/P!=NP/YM/Frankl/NS/AC/Clay-closed claims.